James Yihan Zhang

thejameszhang.github.io | 617-691-9580 | james.y.zhang@yale.edu | LinkedIn | Github

Education

Yale University — Sep 2025 – Present
Visiting Graduate Student: Financial Economics I, Econometrics I (First-Year PhD Courses) — *Grades: Honors*

University of Maryland, College Park — Aug 2022 – May 2025
B.S. Computer Science & Mathematics, Minor Computational Finance — *GPA: 3.9/4.0*

Experience

Yale University — New Haven, Connecticut
Pre-Doctoral Research Fellow, School of Management — *Aug 2025 – Present*
with Bryan Kelly, Head of Machine Learning at AQR

- Building a large language model–based annotation pipeline that reads Dow Jones Newswires articles, produces structured belief annotations, and assigns each story to the relevant global futures markets for empirical finance applications.
- Cutting GPU costs on the Yale high-performance computing cluster by 3x across 16 million vLLM inference calls to Gemma4 26B-A4B via prefill caching optimizations and recall-tuned keyword filtering.
- Extending the text-as-data finance literature by showing that ridge forecasts using news signals earn 9.7% annual alpha (t=3.0) on 25 commodity futures, surviving controls for momentum, value, basis, and basis-momentum factors.
- Building and publishing Global Macroeconomic Data, a Python package that builds comprehensive datasets covering returns of 200 global futures markets using data from LSEG Tick History, Datastream, WRDS, Compustat, OECD, and FRED.
- Developing a dynamic macroeconomic knowledge graph built from financial news to investigate Robert Shiller's "excess volatility" puzzle, linking the evolving flow of macro information to asset prices.

Independent Trading – Kalshi — New Haven, Connecticut
Proprietary Prediction Markets Trader — *Jan 2026 – Present*

- Generated a 19% year-to-date net return on $14,000 of personal capital with an annualized Sharpe ratio of 2.2.
- Deploying automated market making systems in Rust on an AWS Lightsail instance trading across WTI crude oil, Bitcoin, and other daily markets with real-time position tracking and risk controls.
- Using live CME Globex MDP 3.0 options data from Databento to implement pricing, volatility modeling, and delta hedging.
- Building a sportsbook that quotes same-game parlays on Kalshi, pricing correlated legs off FanDuel's parlay odds and quoting lines into live request-for-quote auctions to capture the vig across NBA, WNBA, MLB, and World Cup matches.

Amazon — Seattle, Washington
Software Engineering Intern — *Jun 2025 – Aug 2025*

- Spearheaded a real-time performance analytics project in Java, presenting key metrics to over 2 million active Amazon Sellers; led a database migration from Elasticsearch to AWS Timestream, projected to cut costs by 10% and user latency by 5%.
- Received a full-time return offer for the Amazon Video Shopping Experiences team.

University of Maryland, College Park — College Park, Maryland
Research Assistant, Department of Finance — *Jan 2024 – May 2025*
with Serhiy Kozak

- Designed a class of latent, three-dimensional factor models ("Tensor Factor Models") motivated by tensor decompositions, capturing drivers of asset returns across time, lags, and cross-sectional dimensions; implemented the model in Python using a dataset of 96 monthly factor portfolios.
- Wrote GPU-accelerated Orthogonalized Alternating Least Squares and PARAFAC algorithms in Jax, enabling scalable factorization of high-dimensional firm characteristics data.
- Implemented a kernel-methods model of the cross-section of returns, writing a custom Gaussian-kernel operation in C++/CUDA and registering it as a native, differentiable Jax primitive (JIT, vmap, autodiff) for GPU-scale estimation.

Capital One — McLean, Virginia
Software Engineering Intern — *Jun 2024 – Aug 2024*

- Built a Slackbot using FastAPI for the Data Remediation team, supporting Capital One's internal workflows and enabling faster resolution tracking on data lake platforms; improved query speeds by 15% and cut costs by 5%.

Profile

Technical Skills: Python, Rust, Java, C++, CUDA, Polars, Jax, PyTorch, Huggingface, vLLM, NumPy, Pandas, LaTeX
Teaching Experience: Teaching Assistant for BUFN402 (Portfolio Management)
Professional Development: Yale School of Management Weekly Seminars in Finance, Yale Tobin Center Weekly Pre-Doctoral Seminars in Economics, Yale Empirical Asset Pricing Reading Group
Research Interests: Financial Machine Learning, Global Macro, Commodity Markets, Prediction Markets, Large Language Models, Market Microstructure, Random Matrix Theory